

Day-6 Wireless Standard & Configuration : Wireless Setup, Channel Selection & Avoiding Interference, Wireless Security, MAC Filtering & Access Control Lists

Task 1.

Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

IP assignment:	Automatic (DHCP)
DNS server assignment:	Automatic (DHCP)
SSID:	POCO X3
Protocol:	Wi-Fi 4 (802.11n)
Security type:	WPA2-Personal
Manufacturer:	Realtek Semiconductor Corp.
Description:	Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
Driver version:	6001.15.160.101
Network band (channel):	2.4 GHz (6)
Aggregated link speed (Receive/Transmit):	144/144 (Mbps)
IPv4 address:	10.254.204.156
IPv4 default gateway:	10.254.204.135
IPv4 DNS servers:	10.254.204.135 (Unencrypted)
Physical address (MAC):	60:FF:9E:4F:CE:82

Task 2.

Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks.

Hint: Look for overlapping channels and pick one with the least overlap.

The common 2.4 GHz non-overlapping channels are 1, 6, and 11.

1. Install a Wi-Fi analyzer app on your Android phone, such as a Wi-Fi analyzer.
2. Open the app near your router.
3. Check nearby Wi-Fi networks.
4. Look at channels 1, 6, and 11.
5. Select the channel with the least interference/congestion.
6. Log in to your router.
7. Go to Wireless → 2.4 GHz → Channel.
8. Change Auto to your selected channel.
9. For example, select Channel 6.
10. Save/apply the settings.
11. Reconnect your phone/laptop to Wi-Fi.
12. Check the analyzer again.

Task 3.

Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

1. Log in to your router.
2. Go to Wireless → Security or Wi-Fi Security.
3. Find Security Mode.
4. Select:
 - WPA2-Personal (WPA2-PSK), or
 - WPA3-Personal, if your router and devices support it.
5. Enter a strong Wi-Fi password.

Example of a strong password:

Ketan@Net2026#WiFi

6. Click Save/Apply.

7. Your Wi-Fi may disconnect temporarily.
8. On your phone/laptop, select the Wi-Fi network again.
9. Enter the new password.
10. Verify that it connects successfully.
11. Try the old password and verify that it does not connect.

Task 4.

Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result.
Hint: You can find your device's MAC address in the network settings.

MAC filtering allows the router to permit or deny devices based on their MAC addresses.

First, find your device's MAC address

On Android, generally:

Settings → About Phone → Status → Wi-Fi MAC Address

On Windows:

1. Press Windows + R.
2. Type cmd.
3. Press Enter.
4. Run:

ipconfig /all

5. Look for Physical Address under your Wi-Fi adapter.

Example:

A4-B1-C2-D3-E4-F5

Configure the router

1. Log in to the router.
2. Find MAC Filtering / Access Control / Wireless MAC Filtering.
3. Enable it.

4. Select Allow/Whitelist mode.
5. Add your device's MAC address.
6. Save/apply the settings.
7. Verify that your device can connect.

Now take a second device whose MAC address has NOT been added.

8. Try connecting the second device to the Wi-Fi.
9. It should be blocked if whitelist mode is configured correctly.

Device	MAC Added?	Result
Laptop	Yes	Connected
Phone/Other device	No	Blocked

Task 5.

Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.

This task depends heavily on your router model. Home routers may call this feature Access Control, Website Filtering, URL Filtering, Parental Controls, or Firewall Rules rather than "ACL."

1. Log in to your router.
2. Go to Advanced → Security / Access Control / Parental Controls.
3. Enable the access-control feature.
4. Create a new rule.
5. Select your Wi-Fi device or LAN clients.
6. Choose Block/Deny.
7. Add the website/domain:

www.youtube.com

8. Save/apply the rule.
9. On the connected device, close and reopen the browser.
10. Try accessing YouTube.
11. Record whether access is blocked.

Test	Expected Result
Normal website	Accessible
YouTube	Blocked
Wi-Fi connection	Still connected